

AI For Games Demo

“Exploring adaptive Enemy Behaviors through AI communication”

Team Member:

Teng Shi Heng

Goh Jun Jie

Johnny Yong Jun Siang

Exploring adaptive Enemy Behaviors through AI communication

- **Key research focus:**
 - AI Communication:
 - Investigating how AIs share and act on known information
 - Vision based information gathering
 - Using viewcones, gaussian raycasting, radius or poisson disc sampling
 - Sound based information gathering
 - Simulate hearing with dynamic sound propagation and suspicion levels

Goal: Creating an AI that feels autonomous, adaptive

Adaptive AI matters because it enhances immersion

Why does this matters to Game Developers?

- Enhances in game immersion with a human-like enemy behavior
- Modular & Scalable blackboard design for AIs
- Real-time adaptability reduces reliance on scripted behaviors

Example of games that could benefit from this:

- **Halo**
 - Dynamic Ally / Enemy collaboration
- **F.E.A.R**
 - Tactical Enemy Positioning

AI detects critical information to make Human-Like decision

What information does the Enemy AI store / detects for?

- Last known player position
- Suspicious Sound data
- Known health pack locations

As a result:

AI adapts to player actions in real time, thus allowing it to be

DYNAMIC

Case 1: Enemy sees all, an omniscient entity

- No exploration needed
- AI has full awareness of all map relevant data (Player positions, Health pack positions)

Case 2: Basic circular proximity gives AI limited but immediate awareness

- Enemy are able to detect any information that happen within their radius
 - Depicted in yellow when not detecting the player
 - Gradually turns to red when the player is within their radius



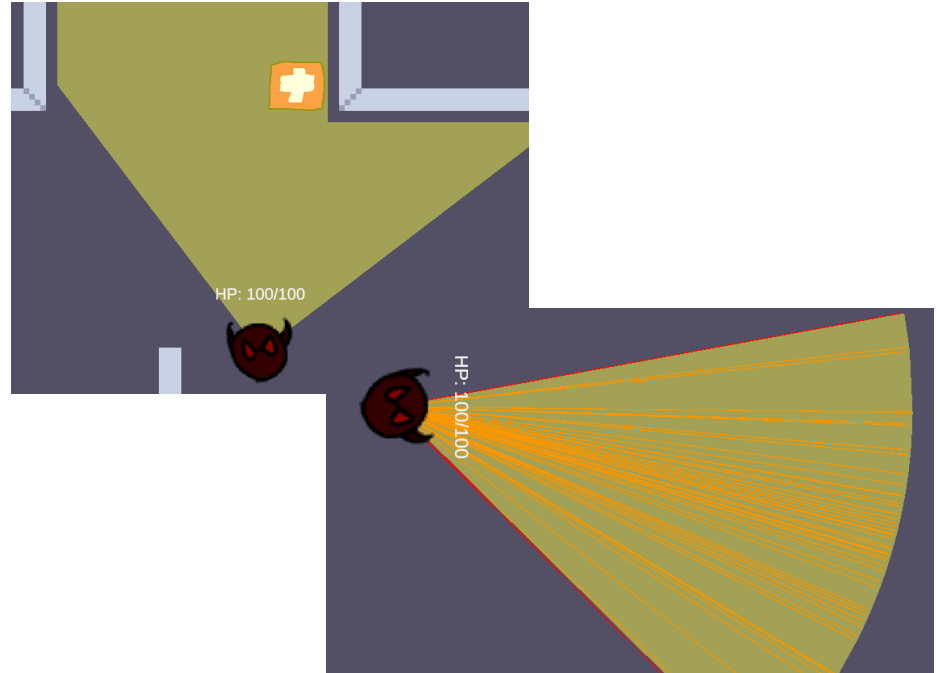
Case 3: View Cone with Gaussian Raycast mimics human vision

A simple view cone:

- More directional based
- Similar to radius detection, but in a cone like shaped instead

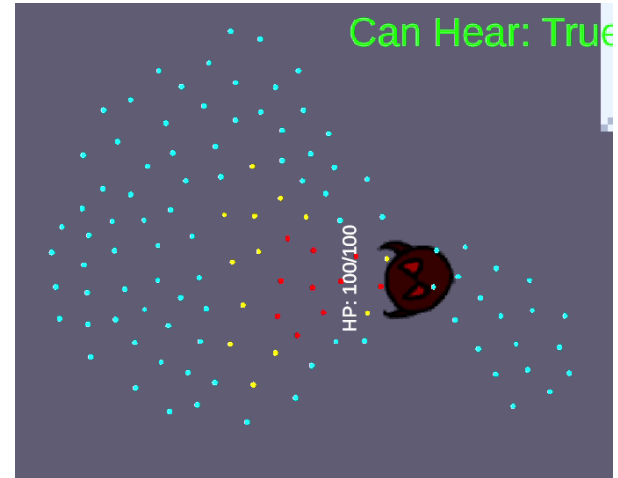
With Gaussian Randomness:

- Allows for bell-curve raycast distribution
- Center rays are visually denser, while outer rays are sparser
- Discovery: Flicker due to the updated bell curve calculated every frame



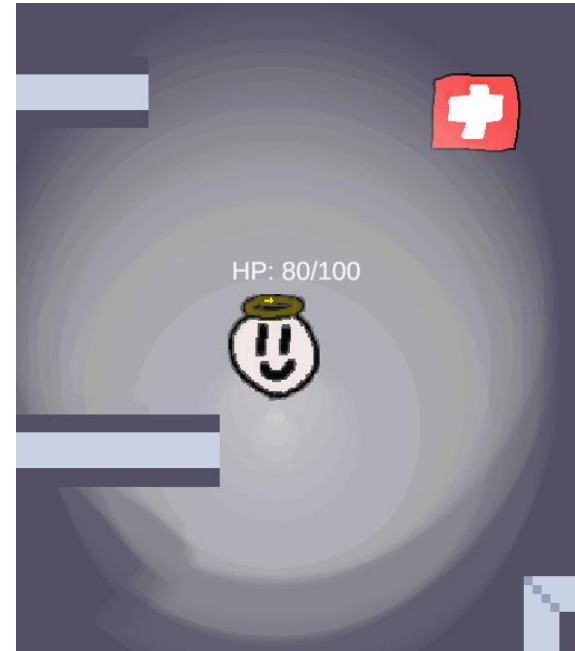
Case 4: Poisson Disc Sampling, a simulation of visual attention

- Uniform, Natural Sampling
 - Ensures no overlapping or clustered vision points
 - Dynamic:
 - Scales sample count during alert vs patrol state
- Recognition-based detection
 - Represent by the dots
 - **Red** (>0.8): High certainty
 - **Yellow** (~ 0.5): Mid certainty
 - **Cyan** (~ 0.2): Low certainty
- Vision
 - Forward Vision
 - Backward Vision
 - To simulate subtle back-sensing



Case 5: AI hearing simulates human-like suspicion

- Sounds are emitted by
 - Player Movement
 - Bullet Movement
- Visualised as expanding ripples
 - White circles growing from the source
- Effective volume decreases with distance & wall
 - Muffled sound when the sound got blocked by a wall
- Has a recognition threshold
 - Suspicion level will be raised temporarily when a sound is heard, based on the sound's volume
 - Simulates thinking process



Action: AI acts rationally based on its knowledge

When the player position is detected:

- If the current health of the enemy is healthy
 - Begins to chase after the player
 - Path calculated by **A*** Algorithm

When the enemy health is below threshold (<50%)

- If a recognised health pack location is known
 - Prioritise reaching the health pack position

When a sound is heard:

- Suspicion gets raised if it is within sound radius
 - Walks towards the sound origin if volume is high enough



AI Communication

Humanistic Way of Communication

Decentralized Memory

- Each AI uses its own local blackboard to track information

Proximity-Based Sync

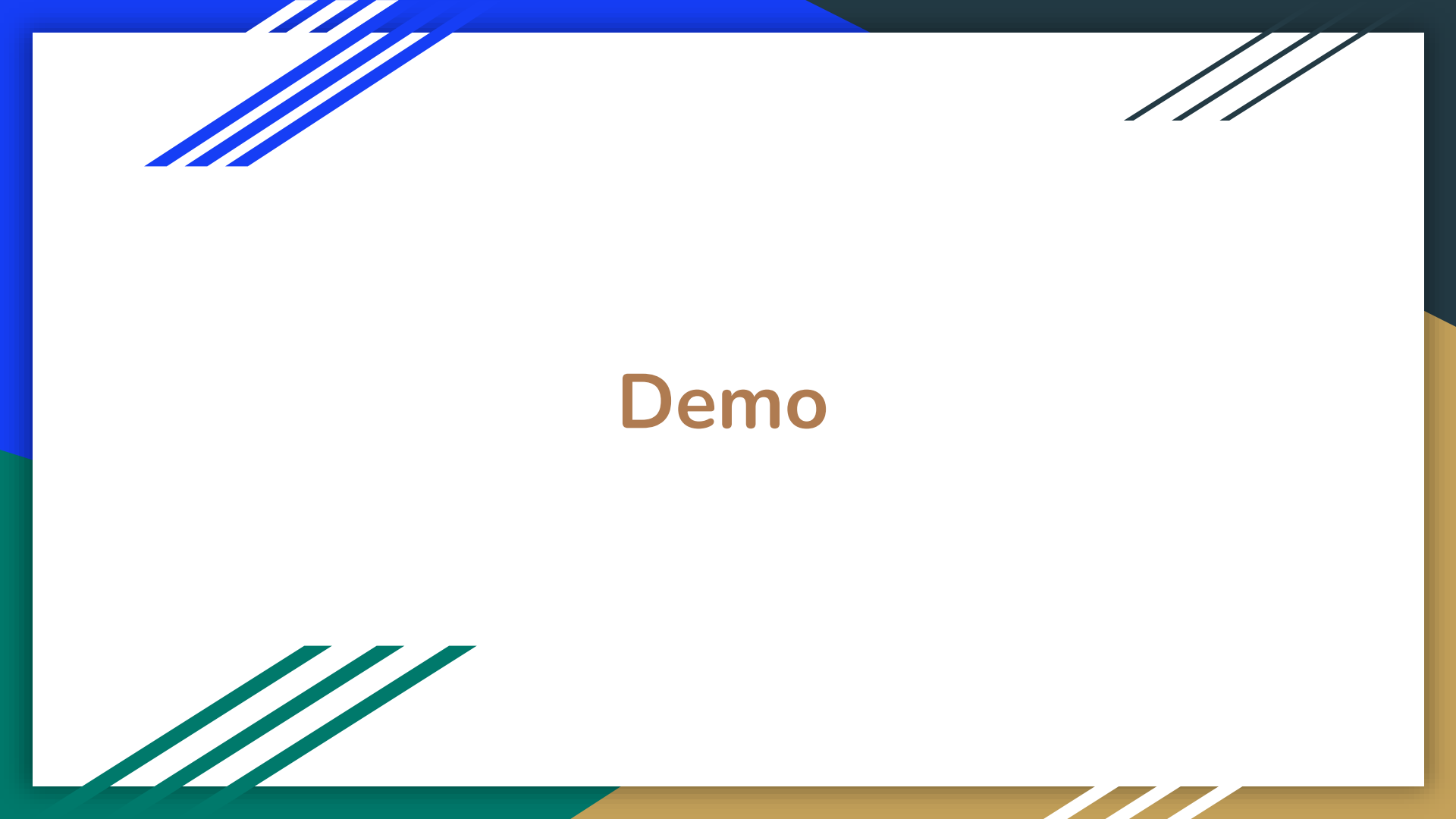
- Agents only share data when they come within detection range of each other

Human-Like Sharing

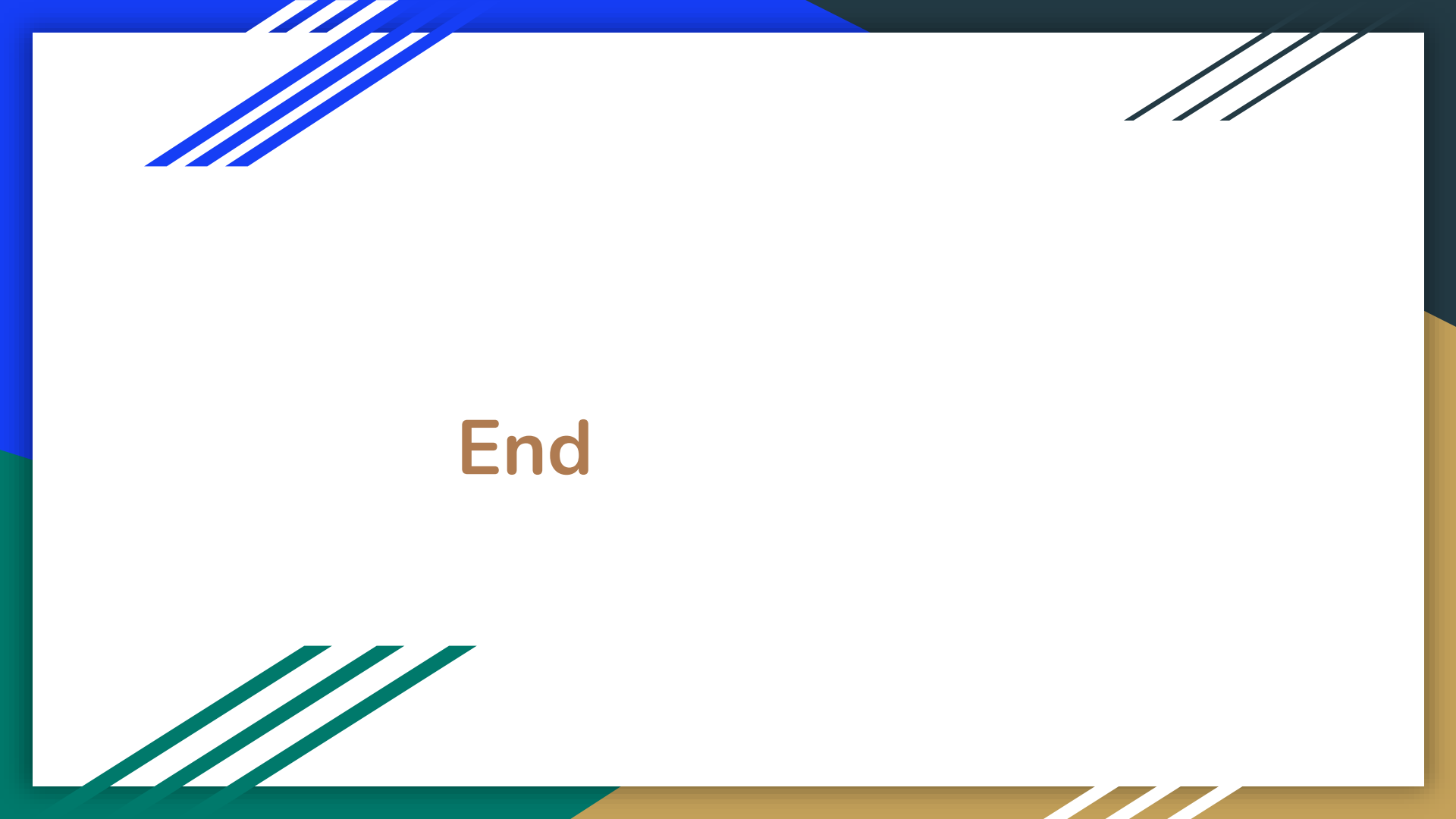
- On detection, AIs compare timestamps of known info and sync based on newest data

Emergent Knowledge Propagation

- Information spreads through encounters, just like human conversations



Demo



End